



Project Summary: Sales Analysis

I have completed a data analysis project on sales data using a dataset compiled from 12 months of retail transactions.

The objective of this project was to identify **key trends, best-performing months, cities, and products**, and provide insights into customer purchasing behavior. This project's key steps were data cleaning, merging, feature engineering, and visual analysis.



What I Analyzed:

- Combined 12 monthly CSV files into one master dataset
 - Cleaned the data by removing NaNs and unnecessary rows
 - Created new columns like **Month, City, Sales, and Hour**
 - Analyzed best months, top cities, and best-selling products
 - Studied time-based patterns to suggest optimal ad display times
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Key Insights and Findings:

1. 📅 Best Month for Sales

- **December** had the highest total sales.
- The holiday season boosts product demand.

2. 🏙️ Top City for Sales

- **Austin** generated the highest revenue.
- Followed by Seattle and Atlanta City.

3. 🕒 Best Time to Display Ads

- Sales peak at **11 AM** and **7 PM**.
- These times are ideal for online ad campaigns.

4. 📦 Best-Selling Products

- Products with **lower prices** sold more units.

✅ Final Conclusion

To boost sales and customer engagement, businesses should:

- Focus marketing during peak hours (11 AM and 7 PM)
- Promote products that complement top sellers
- Use city-specific sales data to create local strategies
- Plan big campaigns for December to make the most impact

This project shows how using data can help make better business decisions.